Modern Academy
for Engineering and Technology in Maadi
Electronic Engineering and Communications
Technology Department

Academic Year	2018-2019
Semester	First
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QUESTIONS FOR THE FINAL WRITTEN EXAM

Subject:	(E501) Digital Signal Processing	**Spec.:**	5th Elec. Eng. & Comm. Tech. Dep.
Examiners:	Dr. Samir Kamal and Dr. Sara Hassan	**Time:**	3 hours

Number of Pages: 2 | Number of Questions: 5 | Attempt all questions

Question 1 (30 points)

a - In detail, what are the main characteristics of the discrete time sinusoidal signals?

b- Check the periodicity of the following sequences and find their fundamental periods.

(i) X (n) = 2 Sin (3 π n / 105). (ii) x (t) = cos (3000 π t).

(c) For the signal $x(t) = 10\cos(1500\pi t)$:

(i) If it is sampled with F_s = 3000 sample/sec, is it possible to reconstruct this signal back from its samples successfully?

(ii) Find the Nyquist frequency, F_N, for this signal.

(iii) Using F_S, find an expression for the corresponding sequence, x(n).

(iv) Find the values of x(n) in the first cycle.

(v) Plot (showing the steps) the given $x(t)$ and the obtained x(n) on the same figure.

Question 2 (20 points)

a- Find the relationship between the frequency of analog signal, frequency of discrete-time signal and sampling frequency.

b- Show that the DFT, X(K), of a sequence x(n) is periodic.

c- For the following sequence: x(n) = {1, 0, 1, 1}:

(i) Using the direct method, DFT, find the frequency components of this sequence.

(ii) Check your results obtained in part (i) by calculating x(n) using the IFFT algorithm.

(iii) Using your results, verify Parseval's theorem.

(iv) Plot the amplitude and the phase spectrums.

(v) What is the computational saving if we use FFT algorithm instead of DFT algorithm.

Question 3 (15 points)

a- Define (using equations and/or figures) the following items: Correlation operation, Correlation coefficient and Convolution operation.

b - Given two sets of samples $x_1(n)$ = {3, 2, 0, 1} and $x_2(n)$ = {1, 2, 0, 1}.

(i) Find the linear cross- correlation between $x_1(n)$ and $x_2(n)$.

(ii) Plot the correlation function obtained in part (i) v.s. the lag number, j.

(iii) Find the autocorrelation function of the sequence $x_2(n)$.

c- For the following sequences, $x_1(n)$ = {3, 2, 0, 1} & x_2 (n) = {1, 2, 0, 1}, Find the convolution of them and plot this convolution.

Question 4 (20 points)

a- For LTIS define (using equations and/or figures): Difference equation, Stability, ROC and linearity.

b- For the following transfer function: $H(z) = z^2(z+0.5) / (z-0.5)*(z-1)^2$

(i) Plot the pole - zero diagram of H (z).

(ii) Discuss the stability of this system.

(iii) Find the difference equation of this filter, y (n).

(iv) Represent the filtering operation in a block diagram form,

c- For H(z) given in (b), using residue method find "4" values for the impulse response of this system.

Question 5 (5 points)

Check (✔) for the right statements and (X) for the wrong statements and correct the mistakes.

i- In quantization process, when the quantization step increases, the accuracy of the quantizer increases. ( )

ii- Nyquist frequency (F_N), is the minimum sampling frequency ( )

iii- Fourier transform (FT) is calculated for a non-periodic continuous – time signals. ( )

iv- Discrete Fourier transform (DFT) is obtained for the analog signals. ( )

v- To reconstruct the continuous time signal, x (t), from its discrete time sequence, X(n), the sampling rate, F_S, must be less than the frequency of x(t) signal, f_0. ( )

Good Luck